

# Supplementary Information

## A Ground–Low-Altitude Mobility Manager for Service Reconfiguration under Disruptions

### S1 Detailed experimental outcomes

These tables report the revised numerical analysis. Means preserve the original fixed scenario weights. Intervals use seed blocks, and B0 D60 incorporates the closed-loop duplicate-command correction. Original data and earlier tables remain in the version archive.

Table S1: E1 service outcomes: mean [95% interval].

| Site | Policy | n   | Completion (s)          | Deadline violations (%) | Damaged services/run |
|------|--------|-----|-------------------------|-------------------------|----------------------|
| A    | B0     | 240 | 187.67 [187.67, 187.67] | 33.33 [33.33, 33.33]    | 0.000 [0.000, 0.000] |
| A    | B1     | 240 | 160.78 [159.13, 162.42] | 25.00 [25.00, 25.00]    | 0.500 [0.500, 0.500] |
| A    | B2     | 240 | 147.62 [147.20, 148.04] | 16.67 [16.67, 16.67]    | 0.058 [0.040, 0.077] |
| A    | B4b    | 240 | 146.63 [145.88, 147.38] | 16.67 [16.67, 16.67]    | 0.133 [0.114, 0.153] |
| B    | B0     | 240 | 262.67 [262.67, 262.67] | 50.00 [50.00, 50.00]    | 0.000 [0.000, 0.000] |
| B    | B1     | 240 | 47.70 [46.49, 48.91]    | 0.00 [0.00, 0.00]       | 0.500 [0.500, 0.500] |
| B    | B2     | 240 | 47.70 [46.49, 48.91]    | 0.00 [0.00, 0.00]       | 0.500 [0.500, 0.500] |
| B    | B4b    | 240 | 49.10 [47.03, 51.17]    | 0.00 [0.00, 0.00]       | 0.487 [0.473, 0.502] |
| C    | B0     | 240 | 323.67 [323.67, 323.67] | 83.33 [83.33, 83.33]    | 0.000 [0.000, 0.000] |
| C    | B1     | 240 | 168.28 [165.32, 171.23] | 10.00 [4.12, 15.88]     | 0.500 [0.500, 0.500] |
| C    | B2     | 240 | 168.28 [165.32, 171.23] | 10.00 [4.12, 15.88]     | 0.500 [0.500, 0.500] |
| C    | B4b    | 240 | 168.65 [165.58, 171.72] | 10.00 [4.12, 15.88]     | 0.500 [0.500, 0.500] |

Completion, damage and deadline-rate intervals use 20 seed blocks. Each block averages the same 12 scenarios. Zero-width intervals indicate no seed-to-seed variation in that fixed-panel average; they do not establish a universal rate. Each site–policy cell contains 240 runs.

Table S2: E2 all-run transport outcomes and conditional path restoration.

| Site | Policy | All runs | Completion (s), mean<br>[95% CI] | Deadline violations<br>(%) | Recovered/affected | Restoration (%) |
|------|--------|----------|----------------------------------|----------------------------|--------------------|-----------------|
| A    | B0     | 320      | 182.00 [182.00,<br>182.00]       | 100.00                     | N/A                | N/A             |
| A    | B1     | 320      | 162.00 [162.00,<br>162.00]       | 37.50                      | 200/200            | 100.00          |
| A    | B2     | 320      | 162.00 [162.00,<br>162.00]       | 37.50                      | 200/200            | 100.00          |
| A    | B4b    | 320      | 166.53 [164.41,<br>168.64]       | 39.06                      | 191/191            | 100.00          |
| B    | B0     | 320      | 241.00 [241.00,<br>241.00]       | 100.00                     | N/A                | N/A             |
| B    | B1     | 320      | 153.88 [153.88,<br>153.88]       | 37.50                      | 200/200            | 100.00          |
| B    | B2     | 320      | 153.88 [153.88,<br>153.88]       | 37.50                      | 200/200            | 100.00          |
| B    | B4b    | 320      | 154.05 [153.94,<br>154.16]       | 37.50                      | 199/199            | 100.00          |
| C    | B0     | 320      | 310.00 [310.00,<br>310.00]       | 100.00                     | N/A                | N/A             |
| C    | B1     | 320      | 278.00 [278.00,<br>278.00]       | 62.50                      | 200/200            | 100.00          |
| C    | B2     | 320      | 278.00 [278.00,<br>278.00]       | 62.50                      | 200/200            | 100.00          |
| C    | B4b    | 320      | 278.92 [277.79,<br>280.06]       | 62.81                      | 201/201            | 100.00          |

All-run denominators are 320 per site and policy. Recovery is conditional on actual primary-chain exposure; B0 is N/A. Restoration is reported as observed recovered/affected counts. The fixed-panel dependence is retained without an independent-run Wilson interval.

Table S3: Mean E3 system weighted loss by disturbance/task level.

| Site | Level | Runs/policy | B0      | B1      | B2      | B4b     |
|------|-------|-------------|---------|---------|---------|---------|
| A    | L1    | 40          | 240.000 | 0.000   | 0.000   | 6.000   |
| A    | L2    | 80          | 240.000 | 240.000 | 240.000 | 240.000 |
| A    | L3    | 80          | 240.000 | 0.000   | 0.000   | 1.688   |
| A    | L4    | 40          | 240.000 | 120.000 | 120.000 | 123.375 |
| B    | L1    | 40          | 240.000 | 0.000   | 0.000   | 0.000   |
| B    | L2    | 80          | 240.000 | 120.000 | 120.000 | 120.000 |
| B    | L3    | 80          | 375.000 | 135.000 | 135.000 | 135.000 |
| B    | L4    | 40          | 375.000 | 255.000 | 255.000 | 255.000 |
| C    | L1    | 40          | 240.000 | 240.000 | 240.000 | 240.000 |
| C    | L2    | 80          | 240.000 | 240.000 | 240.000 | 240.000 |
| C    | L3    | 80          | 375.000 | 135.000 | 135.000 | 135.000 |
| C    | L4    | 40          | 375.000 | 375.000 | 375.000 | 375.000 |

L1/L4 have 40 runs and L2/L3 have 80 per policy. Levels change disturbance structure and demand. SWL uses flat configured penalties and priority weights, rather than monetary costs.

Table S4: E4A observation scheduling, completion and B4b input demand.

| Interval (s) | B0<br>completion<br>(s) | B1<br>completion<br>(s) | B2<br>completion<br>(s) | B4b<br>completion<br>(s) | B4b late (%) | B4b deci-<br>sions/run | B4b prompt<br>tokens/run |
|--------------|-------------------------|-------------------------|-------------------------|--------------------------|--------------|------------------------|--------------------------|
| 10           | 192.0                   | 140.0                   | 140.0                   | 140.0                    | 0            | 62                     | 172849.0                 |
| 30           | 212.0                   | 150.0                   | 150.0                   | 150.0                    | 0            | 22                     | 62307.6                  |
| 60           | 253.0                   | 180.0                   | 180.0                   | 180.0                    | 0            | 12                     | 35017.4                  |
| 120          | 253.0                   | 240.0                   | 240.0                   | 240.0                    | 100          | 7                      | 21348.2                  |
| 300          | 482.0                   | 521.0                   | 482.0                   | 482.0                    | 100          | 4                      | 11707.6                  |

Each policy–arm cell contains 20 runs. Decision counts cover the complete scheduled polling horizon, including post-resolution observations. Token means retain recorded backend outcomes.

Table S5: E4B injected execution delay and mean completion duration.

| Delay (s) | B0 (s) | B1 (s) | B2 (s) | B4b (s) | B4b late (%) |
|-----------|--------|--------|--------|---------|--------------|
| 0         | 182.0  | 120.0  | 120.0  | 120.0   | 0            |
| 1         | 183.0  | 121.0  | 121.0  | 121.0   | 0            |
| 5         | 187.0  | 125.0  | 125.0  | 125.0   | 0            |
| 10        | 192.0  | 130.0  | 130.0  | 130.0   | 0            |
| 20        | 202.0  | 140.0  | 140.0  | 140.0   | 0            |
| 30        | 212.0  | 150.0  | 150.0  | 150.0   | 0            |
| 60        | 242.0  | 341.0  | 302.0  | 302.0   | 100          |

Each policy–arm cell contains 20 runs. Waiting times are controlled simulation interventions, not measured inference latency.

Table S6: E4C B4b input burden with an unchanged core fleet.

| Total aircraft records | Core aircraft | Completion (s) | On time (%) | Logged calls (20 runs) | Prompt tokens/logged call |
|------------------------|---------------|----------------|-------------|------------------------|---------------------------|
|                        | 5             | 4 121.5        | 100         | 41                     | 3982.1                    |
|                        | 10            | 4 120.0        | 100         | 40                     | 5920.6                    |
|                        | 20            | 4 120.0        | 100         | 40                     | 9588.7                    |
|                        | 30            | 4 120.0        | 100         | 40                     | 13256.7                   |
|                        | 50            | 4 120.0        | 100         | 40                     | 20593.5                   |

Each arm contains 20 B4b runs. Added records represent unavailable and non-commandable aircraft. Token means include retained error calls with zero recorded tokens; active fleet size stays four.

Table S7: Aggregate E3 outcomes across the 12-scenario panel.

| Site | Policy | n   | Completion (s), mean [95% CI] | Deadline violations (%) | SWL, mean [95% CI]         |
|------|--------|-----|-------------------------------|-------------------------|----------------------------|
| A    | B0     | 240 | 182.00 [182.00, 182.00]       | 100.00                  | 240.000 [240.000, 240.000] |
| A    | B1     | 240 | 187.25 [187.25, 187.25]       | 41.67                   | 100.000 [100.000, 100.000] |
| A    | B2     | 240 | 187.25 [187.25, 187.25]       | 41.67                   | 100.000 [100.000, 100.000] |
| A    | B4b    | 240 | 188.80 [187.18, 190.41]       | 42.08                   | 102.125 [99.578, 104.672]  |
| B    | B0     | 240 | 241.00 [241.00, 241.00]       | 100.00                  | 307.500 [307.500, 307.500] |
| B    | B1     | 240 | 146.83 [146.83, 146.83]       | 25.00                   | 127.500 [127.500, 127.500] |
| B    | B2     | 240 | 146.83 [146.83, 146.83]       | 25.00                   | 127.500 [127.500, 127.500] |
| B    | B4b    | 240 | 147.05 [146.78, 147.32]       | 25.00                   | 127.500 [127.500, 127.500] |
| C    | B0     | 240 | 310.00 [310.00, 310.00]       | 100.00                  | 307.500 [307.500, 307.500] |
| C    | B1     | 240 | 322.67 [322.67, 322.67]       | 66.67                   | 227.500 [227.500, 227.500] |
| C    | B2     | 240 | 322.67 [322.67, 322.67]       | 66.67                   | 227.500 [227.500, 227.500] |
| C    | B4b    | 240 | 323.82 [322.97, 324.68]       | 66.67                   | 227.500 [227.500, 227.500] |

Scenario-seed runs receive equal weight. L2/L3 therefore contribute twice as many runs as L1/L4; equal-level weighting would define a different summary.

## S2 Interface and implementation details

### S2.1 Principal notation

Table S8: Principal notation.

| Symbol                                         | Meaning                                                                           | Unit                                |
|------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------|
| $t, \tau(t), t_e, H$                           | Decision time, observation timestamp, scheduled execution and horizon             | s                                   |
| $S_t, \hat{S}_t$                               | Current joint state and observed state                                            | —                                   |
| $r_m, d_m, c_m$                                | Mission release, deadline and completion                                          | s                                   |
| $\mathcal{I}_t, \mathcal{C}_{m,t}$             | Candidate information and legal subset                                            | —                                   |
| $\hat{\eta}_c, \delta_{\text{exec}}, \sigma_m$ | Remaining completion estimate, execution delay, estimated temporal margin         | s                                   |
| $a^{\text{prop}}, u, Q, \Gamma$                | Proposal, executed intervention, current-version indicator, current-state checker | —                                   |
| $J_t(c)$                                       | Specified one-step heuristic score                                                | Configured seconds-equivalent units |
| $A_i, R_i$                                     | Affected-chain and recovered-path indicators                                      | Binary                              |
| $T_m, V_m, L_H$                                | Completion duration, deadline violation, weighted service loss                    | s; binary; loss units               |

## S2.2 Structured action vocabulary

Table S9: Structured action vocabulary.

| Action          | Meaning                                                   |
|-----------------|-----------------------------------------------------------|
| DISPATCH        | Assign an available compatible aircraft                   |
| REASSIGN        | Transfer an eligible occupied aircraft to another mission |
| REROUTE         | Change an ongoing operation’s route                       |
| DIVERT          | Change an air operation’s destination or continuation     |
| DELAY           | Postpone an operation                                     |
| CANCEL          | Cancel a specified operation                              |
| RESERVE         | Reserve a resource for a mission                          |
| RETURN          | Request return by an eligible aircraft                    |
| LAND            | Request landing under modeled constraints                 |
| GROUND_FALLBACK | Establish ground service for an eligible mission          |
| NO_ACTION       | Make no operational change at the decision epoch          |
| ESCALATE        | Report a need beyond the current action choice            |

The vocabulary is larger than the set exercised by any single scenario. Supporting an action type in the contract does not imply that the experiments independently validate every possible use of that action. The candidate interface retains legal flags and rejection reasons, while the checker applies action-specific conditions.

## S2.3 Policy inputs and parameters

The complete template is supplied as `reproducibility/prompts/manager_v2.txt`; the input renderer and action schema accompany it. B4a and B4b both use the full global snapshot excluding the `candidates` field. B4b appends that field as a separate JSON section, while B4a omits it. Both retain raw resource state, mission assignments, priorities, positions, deadlines, ground ETA, prompt rules and the common checker. Only B4b receives the derived air ETA, completion prediction, per-candidate legality/rejection reason and explicit preempted-mission mapping. The ablation therefore measures explicit candidate-information support, jointly including information derivation and presentation.

Table S10: Input-field comparison for the E1 ablation.

| Field                                                   | B4a      | B4b      |
|---------------------------------------------------------|----------|----------|
| Raw aircraft and mission state; ground ETA              | Included | Included |
| Rules, objectives, output schema, checker               | Same     | Same     |
| Derived candidate air ETA and completion time           | Omitted  | Included |
| Per-candidate legal flag and rejection reason           | Omitted  | Included |
| Predicted deadline violation and preemption consequence | Omitted  | Included |

Table S11: Key operational and analysis parameters.

| Parameter                                  | Value or implemented rule                                                                        |
|--------------------------------------------|--------------------------------------------------------------------------------------------------|
| Core fleet                                 | Two medical UAVs, one logistics UAV, one passenger eVTOL                                         |
| Cruise speed                               | Medical/logistics 15 m/s; passenger 40 m/s; inspection role 10 m/s (no core inspection aircraft) |
| Nominal altitude                           | UAV 328 ft; passenger eVTOL 656 ft                                                               |
| Arrival radius                             | 100 m; this can make realized duration shorter than geometric ETA                                |
| Dispatch overhead                          | 0 s in candidate estimates                                                                       |
| Endurance                                  | Initial 1,800 s, reduced by a seeded 0–15%; candidate reserve margin 60 s                        |
| Busy-aircraft progress                     | Seeded fraction 0.05–0.40 of its initial route                                                   |
| Simulation step / ordinary decision period | 1 s / 30 s while actionable; E4A overrides observation polling                                   |
| Horizon / emergency release                | 900 s / 300 s                                                                                    |
| E1 deadline slack                          | HIGH 300 s; CRITICAL 180 s                                                                       |
| E2/E4 primary deadline                     | 480 s at Site A; site-adapted configurations accompany the data                                  |
| Incumbent default deadline                 | 1,800 s for initial background missions                                                          |
| Ground completion                          | Execution-time SUMO D1–H1 ETA rounded to seconds, added to issue time                            |
| LLM decoding                               | Temperature 0; JSON output; 8,192 output-token budget                                            |
| SWL priority weights                       | CRITICAL/HIGH/NORMAL/LOW: 4/3/2/1; penalties are task-specific flat values                       |

B2 weights and the prompt template are unchanged. Mission-specific delay and cancellation penalties, secondary releases, workload assignments and site adaptations are specified in the supplied YAML matrices. The policy identifier records the configured backend service; no unverified underlying model-version claim is made.

## S3 Statistical and outcome conventions

### S3.1 Fixed-panel estimand and random streams

The runner initializes a fresh `random.Random(seed)` for every scenario. Aircraft are visited in sorted order; every aircraft consumes an endurance draw and each busy aircraft an additional progress draw. The same SUMO seed is also reused. Workload-dependent draw consumption changes realizations, but does not create independently keyed scenario streams. We retain dependence within each seed across the fixed scenario panel and pair policies by scenario and seed. This choice follows implementation, not seed labels or state hashes alone.

For a complete panel, each seed contributes the average of its scenario-specific outcomes or paired differences. Means retain equal scenario–run weights. A  $t$  interval uses the standard deviation across 20 seed averages (10 in the E1 ablation). For a conditional endpoint with  $N$  retained pairs, let  $\bar{d}$  be the original pair-weighted mean and  $K$  the number of represented seeds. The cluster contribution is

$$B_s = \bar{d} + \frac{K}{N} \sum_{i: s_i=s} (d_i - \bar{d}).$$

The interval is  $\bar{d} \pm t_{.975, K-1} \text{sd}(B_s) / \sqrt{K}$ . This preserves the jointly affected pair-weighted estimand instead of silently reweighting seeds with different exposure counts. Conditional populations remain policy-dependent: at Site C, B4b has 201 affected runs and the B4b–B2 recovery comparison retains 200 jointly affected pairs.

Continuous tests use these seed contributions with the original constant-difference signed-rank fall-back. Binary tests enumerate policy-label swaps of whole seed blocks; the test statistic is the total discordant paired difference. This retains the paired design while replacing an independent-pair McNemar assumption. No scenarios, endpoints or correction families are selected according to their p-values. Intervals are unadjusted; Holm-adjusted p-values retain the original three E1 ablation, 12 E2 and 12

E3 compound endpoints per site. E1 scenario-specific and E4 single-scenario comparisons retain their original seed units. The E2 family includes completion, recovery time, replan latency, incumbent damage, candidate reduction, deadline violation, restoration, recovered completion, ground fallback, failure-induced fallback, necessary-fallback correctness and air intervention. E3 compares B4b with B1/B2 at L2–L4 for SWL and CRITICAL+HIGH loss.

Table S12: Complete E2 seed-block comparison family, Site A. Differences are B4b minus B2; binary differences are fractions.

| oprule Site | Endpoint            | Pairs | Difference | 95% interval     | Holm p   |
|-------------|---------------------|-------|------------|------------------|----------|
| A           | Completion          | 320   | 4.525      | [2.412, 6.638]   | 0.003069 |
| A           | Recovery time       | 191   | 3.927      | [0.930, 6.924]   | 0.142465 |
| A           | Replan latency      | 191   | 3.927      | [0.930, 6.924]   | 0.142465 |
| A           | Damage              | 320   | 0.003      | [-0.012, 0.018]  | 1.000    |
| A           | Candidate reduction | 320   | 0.013      | [0.000, 0.025]   | 0.336690 |
| A           | Deadline risk       | 320   | 0.016      | [0.003, 0.029]   | 0.437500 |
| A           | Restoration         | 191   | 0.000      | [0.000, 0.000]   | 1.000    |
| A           | Recovered/completed | 191   | 0.000      | [0.000, 0.000]   | 1.000    |
| A           | Ground fallback     | 320   | 0.013      | [0.000, 0.025]   | 0.625000 |
| A           | Failure fallback    | 320   | 0.013      | [0.000, 0.025]   | 0.625000 |
| A           | Necessary fallback  | 320   | -0.016     | [-0.029, -0.003] | 0.437500 |
| A           | Air intervention    | 320   | -0.028     | [-0.048, -0.008] | 0.142465 |

Table S13: Complete E2 seed-block comparison family, Site B. Differences are B4b minus B2; binary differences are fractions.

| oprule Site | Endpoint            | Pairs | Difference | 95% interval    | Holm p   |
|-------------|---------------------|-------|------------|-----------------|----------|
| B           | Completion          | 320   | 0.175      | [0.060, 0.290]  | 0.055710 |
| B           | Recovery time       | 199   | 0.281      | [0.098, 0.465]  | 0.055710 |
| B           | Replan latency      | 199   | 0.281      | [0.098, 0.465]  | 0.055710 |
| B           | Damage              | 320   | 0.000      | [0.000, 0.000]  | 1.000    |
| B           | Candidate reduction | 320   | 0.000      | [0.000, 0.000]  | 1.000    |
| B           | Deadline risk       | 320   | 0.000      | [0.000, 0.000]  | 1.000    |
| B           | Restoration         | 199   | 0.000      | [0.000, 0.000]  | 1.000    |
| B           | Recovered/completed | 199   | 0.000      | [0.000, 0.000]  | 1.000    |
| B           | Ground fallback     | 320   | 0.000      | [0.000, 0.000]  | 1.000    |
| B           | Failure fallback    | 320   | 0.000      | [0.000, 0.000]  | 1.000    |
| B           | Necessary fallback  | 320   | -0.003     | [-0.010, 0.003] | 1.000    |
| B           | Air intervention    | 320   | -0.003     | [-0.010, 0.003] | 1.000    |

Table S14: Complete E2 seed-block comparison family, Site C. Differences are B4b minus B2; binary differences are fractions.

| oprule Site | Endpoint            | Pairs | Difference | 95% interval    | Holm p |
|-------------|---------------------|-------|------------|-----------------|--------|
| C           | Completion          | 320   | 0.922      | [-0.214, 2.057] | 1.000  |
| C           | Recovery time       | 200   | 0.255      | [-0.138, 0.648] | 1.000  |
| C           | Replan latency      | 200   | 0.255      | [-0.138, 0.648] | 1.000  |
| C           | Damage              | 320   | 0.000      | [0.000, 0.000]  | 1.000  |
| C           | Candidate reduction | 320   | 0.000      | [0.000, 0.000]  | 1.000  |
| C           | Deadline risk       | 320   | 0.003      | [-0.003, 0.010] | 1.000  |
| C           | Restoration         | 200   | 0.000      | [0.000, 0.000]  | 1.000  |
| C           | Recovered/completed | 200   | 0.000      | [0.000, 0.000]  | 1.000  |
| C           | Ground fallback     | 320   | 0.000      | [0.000, 0.000]  | 1.000  |
| C           | Failure fallback    | 320   | 0.000      | [0.000, 0.000]  | 1.000  |
| C           | Necessary fallback  | 320   | 0.000      | [0.000, 0.000]  | 1.000  |
| C           | Air intervention    | 320   | 0.000      | [0.000, 0.000]  | 1.000  |

Table S15: E3 compound SWL comparisons. CRITICAL+HIGH differences and tests are identical in these cells; both endpoints remain in each 12-test family.

| oprule Site | Level | Base | Pairs | Difference | 95% interval     | Holm p |
|-------------|-------|------|-------|------------|------------------|--------|
| A           | L2    | B1   | 80    | 0.000      | [0.000, 0.000]   | 1.000  |
| A           | L3    | B1   | 80    | 1.688      | [-1.844, 5.219]  | 1.000  |
| A           | L4    | B1   | 40    | 3.375      | [-3.689, 10.439] | 1.000  |
| A           | L2    | B2   | 80    | 0.000      | [0.000, 0.000]   | 1.000  |
| A           | L3    | B2   | 80    | 1.688      | [-1.844, 5.219]  | 1.000  |
| A           | L4    | B2   | 40    | 3.375      | [-3.689, 10.439] | 1.000  |
| B           | L2    | B1   | 80    | 0.000      | [0.000, 0.000]   | 1.000  |
| B           | L3    | B1   | 80    | 0.000      | [0.000, 0.000]   | 1.000  |
| B           | L4    | B1   | 40    | 0.000      | [0.000, 0.000]   | 1.000  |
| B           | L2    | B2   | 80    | 0.000      | [0.000, 0.000]   | 1.000  |
| B           | L3    | B2   | 80    | 0.000      | [0.000, 0.000]   | 1.000  |
| B           | L4    | B2   | 40    | 0.000      | [0.000, 0.000]   | 1.000  |
| C           | L2    | B1   | 80    | 0.000      | [0.000, 0.000]   | 1.000  |
| C           | L3    | B1   | 80    | 0.000      | [0.000, 0.000]   | 1.000  |
| C           | L4    | B1   | 40    | 0.000      | [0.000, 0.000]   | 1.000  |
| C           | L2    | B2   | 80    | 0.000      | [0.000, 0.000]   | 1.000  |
| C           | L3    | B2   | 80    | 0.000      | [0.000, 0.000]   | 1.000  |
| C           | L4    | B2   | 40    | 0.000      | [0.000, 0.000]   | 1.000  |

### S3.2 Service-loss audit

All 2,880 E3 run registries were checked against their 900 s horizons. At each site, 1,440 task instances had deadlines at or before the horizon, and all were COMPLETED. Across sites, zero overdue tasks remained EN\_ROUTE or ASSIGNED. Thus no SWL correction or overdue-task penalty sensitivity is needed for this dataset. Background missions with 1,800 s deadlines are not treated as failures merely for remaining in service at 900 s. Incumbent damage remains a count of damaged final states. As a complementary E1 measure, the recorded preemption time was linked to the incumbent identified in the candidate log and its final interrupted state. The preemption-to-horizon exposure of these persistently interrupted services averages 35 s per run for B2 and 80 s for B4b at A (B1: 300 s). This is horizon-truncated exposure, not a complete cumulative interruption-cost measure. Mission CSV records omit some incumbent preemption transitions, so zero logged interruption duration would be misleading. Initial incumbent deadlines are 1,800 s, beyond the 900 s horizon; no incumbent lateness claim is inferred from this exposure.

## S4 Technical checks and timing sensitivity

### S4.1 Duplicate pending commands and E4B regression

The original B0 D60 action pipeline records GROUND\_FALLBACK at 300 s, due at 360 s. The 360 s fault invokes a new decision with the identical normalized action and identical 181.763 s ETA field. The original queue replaces the first command, restarts the 60 s wait, issues at 420 s and completes at 602 s. This caused the original 302 s completion duration.

The revision preserves the original due time when a validated normalized action exactly matches a pending one. It continues to supersede a pending command when the action changes, and revalidates the surviving command at execution. The correction is implemented in `queue_fix.py`, as an overlay without altering original project source or raw runs. All 20 B0 D60 seeds were rerun through SUMO-BlueSky: issue at 360 s, completion at 542 s, duration 242 s. Single-seed D00/D30 controls retain 182/212 s; fault times 359/361 s both give 242 s. B2's changed air-to-ground command retains a 302 s duration. B1 retains its air-to-air replacement sequence; patched and unpatched control reruns both give 342 s in the current runtime, versus the archived 341 s, a 1 s runtime reproduction difference unrelated to deduplication. Its archived cell is retained because the correction does not change its command sequence. No LLM calls were made in these regressions; original B4b traces contain genuine command replacement

and remain unchanged.

Table S5, main Figure 6 and Supplementary Figure S2 replace the affected B0 cell only. D60 is an evaluated event-order configuration, not a universal transport-delay threshold. External backend wall time remains separate from controlled simulation waiting.

## S4.2 Physical transfer semantics and offline added-time sensitivity

The model replaces a service assignment; it does not track an individual payload. New air candidates route through the mission origin. Interrupted-service candidates route through V3 before the destination, but a backup can be dispatched before the original carrier has reached V3. Preemption interrupts and detaches the incumbent mission. Ground takeover establishes a service timer using the current reference corridor ETA. There is no explicit unloading, rendezvous, loading, new inventory or re-shipment state.

We therefore evaluate added transfer time as an *offline trajectory-fixed sensitivity*, not a new closed-loop experiment. For an actually affected E2 mission with a post-failure DISPATCH, REASSIGN or GROUND\_FALLBACK, add one hypothesized transfer delay  $h$  to its recorded completion. The same rule applies to every policy; uninterrupted services receive no added time. A previously timely completion changes status when  $0 \leq d_m - c_m < h$ . Choices, fault exposure and later scheduling are held fixed. Values  $h = 15, 30, 60$  s are illustrative assumptions, not measured handling durations.

Table S16: Offline added transfer time: number of previously timely E2 replacements becoming late.

| oprule Site | Policy | Replacements | Timely | +15 s | +30 s | +60 s |
|-------------|--------|--------------|--------|-------|-------|-------|
| A           | B1     | 200          | 80     | 0     | 0     | 0     |
| A           | B2     | 200          | 80     | 0     | 0     | 0     |
| A           | B4b    | 191          | 75     | 1     | 1     | 2     |
| B           | B1     | 200          | 80     | 0     | 80    | 80    |
| B           | B2     | 200          | 80     | 0     | 80    | 80    |
| B           | B4b    | 199          | 80     | 0     | 80    | 80    |
| C           | B1     | 200          | 0      | 0     | 0     | 0     |
| C           | B2     | 200          | 0      | 0     | 0     | 0     |
| C           | B4b    | 201          | 0      | 0     | 0     | 0     |

At Site B, 80 affected replacements per coordinated policy are timely with 19 s remaining; a 30 s addition makes all 80 late. At A, B1/B2 timely replacements have 60 s remaining and none flips for  $h \leq 60$  s; B4b has one flip at 15/30 s and two at 60 s. At C, affected replacements in this subset are already late. For the preemption example in Section 4.2, 17 s of extra air-specific handling erases the realized 212 versus 229 s speed advantage; more than 88 s also loses its emergency deadline. Adding the same handling time to both initial alternatives preserves their speed difference, so this threshold applies specifically to an additional preemption step. These checks show which conclusions depend on omitted stages; they do not verify dynamic rescheduling with payload transfer.

## S4.3 Road-statistics boundary

The original morphology routine summed all imported passenger edges while dividing by  $10.24 \text{ km}^2$ , without clipping to its bbox argument. The revised lengths intersect every edge polyline with the configured geographic rectangle projected through the SUMO network transform. SUMO edge lengths are apportioned by the clipped polyline fraction. Density uses that same polygon's area. Intersection counts retain nodes within the polygon and count distinct neighbors in the full graph, avoiding artificial boundary intersections. Original full-network topology, OD samples and routing remain separately identified and unchanged. Site C still has higher intersection density than A; its description relies on dead ends, degree, circuitry and facility distance.

## S4.4 Fault state machines and recovery conditions

Table S17: Six modeled fault families. Exact targets, geometry, times and windows are supplied in the YAML matrices.

| Fault               | Trigger and state change                                                                    | Independent local response                                                                    | Service-recovery condition                                                                       |
|---------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| F1 C2 loss          | Target becomes non-commandable and in CONTINGENCY; mission interrupted                      | Return to V3                                                                                  | Surviving compatible backup or ground; lost target remains excluded                              |
| F2 GNSS degradation | One-shot test of aircraft inside configured zone; GNSS DEGRADED                             | Occupied affected aircraft abandon mission and divert to V3                                   | Alternative with acceptable navigation and landing state; zone alone does not block other routes |
| F3 UTM change       | DEGRADED blocks new noncritical air assignments; OUTAGE blocks all new air                  | OUTAGE interrupts in-flight missions and diverts them to V3                                   | Ground during outage; otherwise an eligible air assignment                                       |
| F4 Landing closure  | Configured site becomes unavailable; missions to it interrupted                             | Affected aircraft divert to V3                                                                | Available compatible destination/continuation or ground                                          |
| F5 Unknown aircraft | Prescribed noncooperative trajectory and temporary risk zone; affected missions interrupted | Commandable affected aircraft divert to V3                                                    | Aircraft and route outside active zone, or ground; zone expires at configured end                |
| F6 Flyaway          | Target becomes uncontrolled and non-commandable; surrounding envelope can interrupt others  | Target follows prescribed uncontrolled trajectory; other commandable affected aircraft divert | Exclude target and active envelope; use surviving air or ground                                  |

Zone applicability is evaluated at decision and execution, with configured active windows. Communication, navigation, UTM and landing-site states are not automatically repaired during these runs. Recovery denotes establishing replacement service, not repairing the original aircraft or infrastructure.

## S5 Figure production and source records

Figure 1 combines separate, text-free AI-generated illustrations with editable PowerPoint text, routes, containers and markers. Its map is marked Not to scale and its finding graphics are schematic. A later author-edited PowerPoint removed the duplicate figure title and adjusted module labels; these edits are retained in the revision. The visual reference was supplied by the authors; generated illustrations do not depict observations or operational deployments at the study sites. Images were produced with the built-in imagegen tool and composed into PowerPoint. The retained illustration records identify the imagegen tool; no unverifiable model-release version is assigned.

All figure text uses Times New Roman. Main Figure 3 uses native clustered-column charts with the existing seed-block intervals. Main Figure 4 restores the candidate-geometry, preemption-choice and deadline-outcome layout. Main Figure 5 contains editable vector heatmaps and contrasts; main Figure 6 and Supplementary Figure S1 use native charts with embedded workbooks. Supplementary Figure S2 retains editable pending-command and transport segments. The two PowerPoint files retain their source slide order; the editing guide maps slides to current main and supplementary figure numbers.

## S6 Reproduction files and entry points

No public repository URL is assigned. The accompanying REPRODUCTION\_README.md and reproduce\_analysis.ps1 provide original run locations, dependencies and executable processing order. The reproducibility/ folder contains prompts, schemas, weights, experiment configurations, derived data and a hashed file inventory. The local correction overlay and optional regression entry point are included; the regression makes no LLM calls. Editable figure builders and PowerPoint sources accompany the LaTeX package.

## S7 Supporting transport outcomes and command sequence

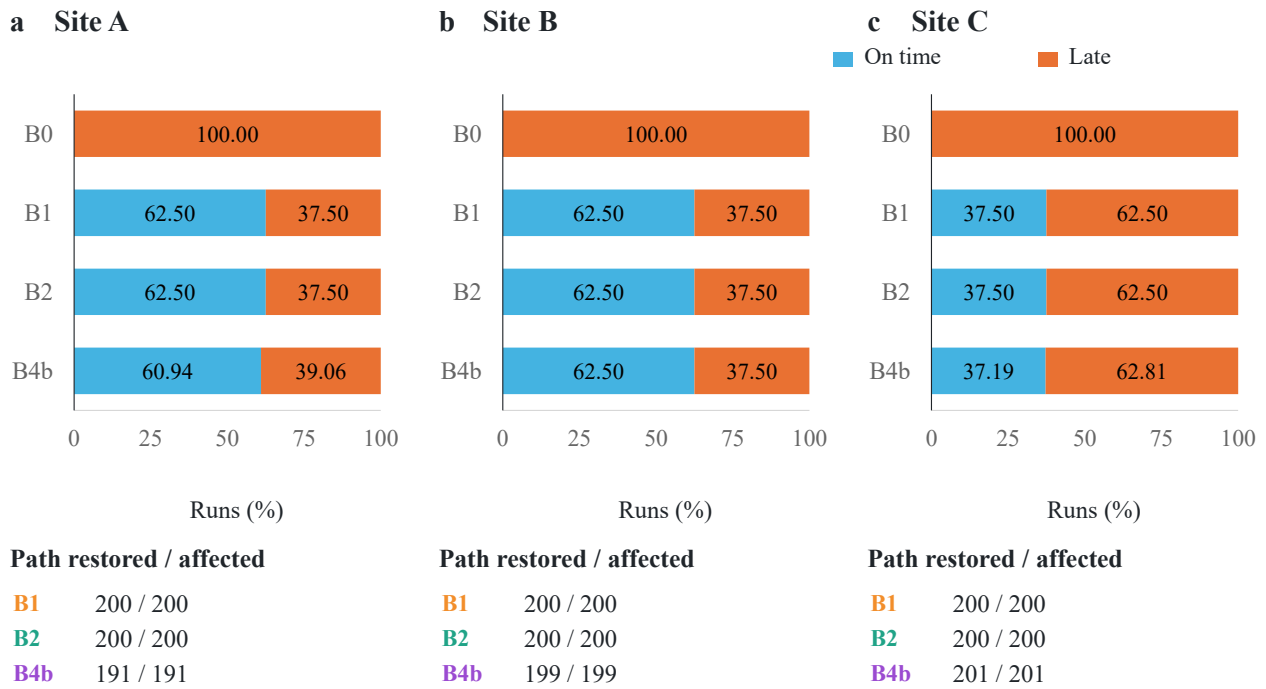


Figure S1: E2 path restoration and timely service answer different questions. (a–c) Bars show on-time and late completion percentages over all 320 runs per site and policy. The separate counts below give restored/actually affected primary air chains for B1/B2/B4b. Recovery denominators can depend on prior policy actions; B0 has no affected-air-chain denominator. Blue and orange encode deadline outcome here, not policy identity. Table S2 retains the full summaries and exact restoration counts.

**a B0: repeated ground command**



**b B4b: air replaced by ground**

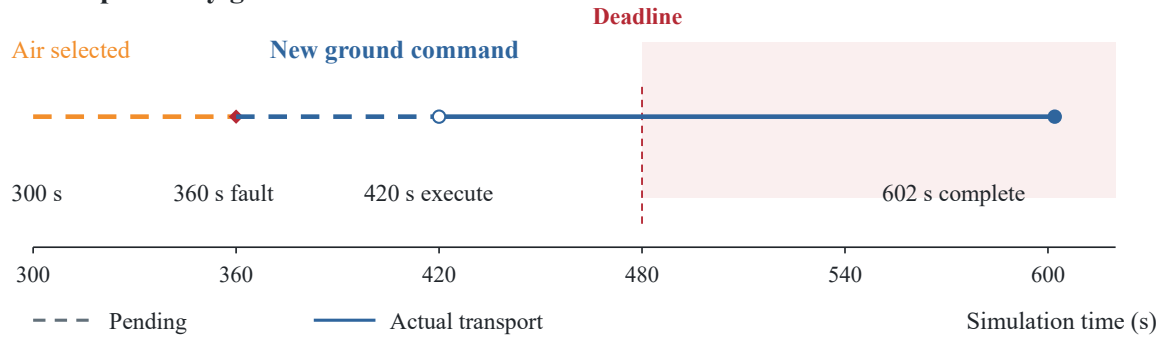


Figure S2: E4B D60 command sequences after identical-command deduplication. The pending air dispatch is superseded by a ground-fallback decision at the fault time. Ground executes at 420 s and completes at 602 s, after the 480 s deadline. Orange denotes the air command, blue the ground service and red the fault/deadline. Dashed lines denote pending commands and solid lines actual transport. Corrected B0 ground service executes at 360 s and completes at 542 s; B4b changes mode and completes at 602 s.